Exact solvability of regular alternating paper-heart patterns

A mathematical note for the Juleflet project

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Abstract

Which binary designs can be reproduced exactly by the regular alternating weave of two slitted sheets? We formalize the visible-boundary and shared-cut construction described by Mogensen. A design is representable precisely when its boundary admits an embedded completion whose two sheet subgraphs are disjoint families of opposite-side paths. We prove this equivalence and derive border-count, junction-degree and closed-component obstructions. Simple local tests are not sufficient: a stripe and a detached disk provide a counterexample. The result concerns ideal planar colour reproduction, not physical assembly, minimum paper width or the completeness of a numerical solver.

1 Model and scope

Rectify the overlap to the square $Q = [0,1]^2$, with sides T, B, L, R. A *regular binary mask* is a colouring M by $\mathbb{F}_2 = \{0,1\}$ of the regions of a finite, embedded, piecewise-smooth boundary graph Γ in Q. Every open edge of Γ separates different colours. Edges have their interiors in $\operatorname{int} Q$; vertices may lie on the sides, but not at corners. There are no boundary edges running along a side, and the colouring is constant near each corner. We identify masks that differ only on their zero-width boundary graphs. This excludes isolated coloured points or lines with no two-dimensional region on either side.

A *legal cut family* $\mathcal{A}$ consists of finitely many mutually disjoint simple arcs, each joining T to B and otherwise lying in $\operatorname{int} Q$. A second family $\mathcal{B}$ joins L to R under the same rules. Empty families are allowed. Within a family, cuts cannot touch, cross, branch, close into loops or stop in the interior. Cuts from different families may cross, touch or coincide along segments. We assume their union has a finite embedded graph subdivision: intersections and shared portions do not accumulate. Neither family is required to be monotone.

The arcs of $\mathcal{A}$ divide Q into ordered strips. Let $p_A(x)$ be the parity of the strip containing x, starting with zero at L; define $p_B(x)$ analogously, starting at B. The regular alternating renderer is

$$W(x) = \phi \oplus p_A(x) \oplus p_B(x), \qquad \phi \in \mathbb{F}_2. \tag{1}$$

Thus crossing one cut changes colour, whereas crossing an exactly shared cut changes both parities and leaves the colour unchanged. This is the construction principle of Mogensen's advanced paper-heart patterns [1, §4]. Other over/under conventions, stopped cuts and removed pieces are outside this model. Equation (1) defines representability here; it does not assert that every such pair can readily be threaded together in real paper.

Lemma 1 (Strip parity)**.** *A legal family of n opposite-side arcs divides the square into $n+1$ ordered strips. Its parity is locally constant away from its cuts and changes across each cut. Endpoints on the two sides occur in corresponding order.*

Proof. A simple proper arc across a disk is a crosscut: closing it by either boundary arc gives a Jordan curve, so it separates the disk into two components. Disjoint crosscuts cannot have alternating endpoints around the boundary, since the first would separate the endpoints of the second. Their endpoints therefore preserve order. Adding the cuts in that order splits one strip at a time, yielding $n+1$ strips. Numbering successive strips gives the claimed parity changes. This argument uses separation, not monotonicity. □

2 The exact completion criterion

Definition 2 (Boundary completion)**.** A completion is a finite embedded graph G in Q containing a subdivision of Γ. Each visible edge, belonging to Γ, is labelled A or B. Every additional edge is labelled H (hidden), with its open interior disjoint from Γ. Write G_A for the subgraph of edges labelled A or H, and G_B for those labelled B or H. Isolated vertices are ignored. A completion is *admissible* when:

(i) In each sheet subgraph, every interior vertex has degree zero or two.

(ii) Boundary vertices of G_A have degree one and lie on T or B; those of G_B have degree one and lie on L or R. No cut meets a corner.

(iii) Every component of G_A is a path from T to B, and every component of G_B is a path from L to R. In particular, there are no cycles or same-side paths.

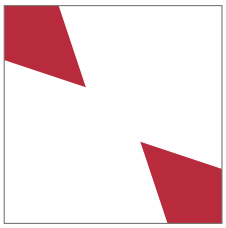
The embedding matters: unintended crossings or contacts must be included as vertices and satisfy these rules. A purely abstract graph with the right degrees is not sufficient.

Theorem 3 (Exact representability)**.** *A regular binary mask is exactly representable by Equation (1), with legal cut families satisfying the finite intersection assumption, if and only if its boundary has an admissible completion. Exactness is understood away from zero-width cut and boundary sets; the phase ϕ is free.*

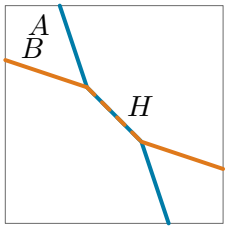
Proof. Necessity. Suppose legal families render M. Subdivide their union at intersections and at endpoints of shared portions. On any open edge used by exactly one family, Lemma 1 and Equation (1) give a colour change. Such edges are exactly the visible boundary; label them by their sheet. An edge used by both families has no colour change; label it H. The original legal arcs are precisely the path components of the resulting sheet subgraphs, so all completion conditions hold.

Sufficiency. Given an admissible completion, take the components of G_A and G_B as the two cut families. Embedded path components are simple, and distinct components of a sheet are disjoint. Their prescribed endpoints make them legal cut families. By Lemma 1, their strip parities are well-defined. Across a visible edge, exactly one parity changes. Across a hidden edge, both change, so their XOR does not. Consequently M and W have the same colour changes across every open edge of G.

Choose ϕ to make $M = W$ at one point outside G. Any other region can be reached by a path avoiding the finitely many vertices and crossing open edges one at a time. At each crossing $M \oplus W$ remains unchanged. Hence $M \oplus W = 0$ in every region, proving exact reproduction. □



(a) Target mask



(b) Admissible completion

Figure 1: One top–bottom cut and one left–right cut share the middle segment. Solid blue and orange segments are visible; the alternating segment belongs to both sheets and is invisible in the mask. At each end of that segment, two visible arms and one shared arm form a legal junction.

3 Consequences and obstructions

Corollary 4 (Boundary counts)**.** *Let t_T, t_B, t_L, t_R count genuine colour transitions along the four sides. Every exact solution satisfies*

$$|\mathcal{A}| = t_T = t_B, \qquad |\mathcal{B}| = t_L = t_R.$$

Moreover, the XOR of the four corner colours is zero.

Proof. Only A-cuts can end on T and B. Each has one distinct endpoint on each side, and each endpoint changes the adjacent strip parity. The same argument applies to B-cuts on L and R. The colour difference between the two top corners is t_T mod 2; the bottom difference is t_B mod 2. Equality of these parities gives the corner assertion. It is a consequence of the count condition, not an additional sufficient test. □

Corollary 5 (Local degree)**.** *At an interior completion vertex, let a, b, h count incident edges labelled A, B, H. Then*

$$a + h \in \{0, 2\}, \qquad b + h \in \{0, 2\}. \tag{2}$$

In particular, the number $d = a + b$ of visible boundary branches is even and at most four. Six or more incident visible branches are impossible.

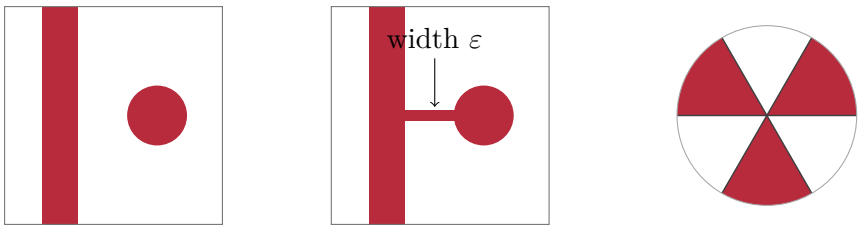
Proof. The two sums are the degrees in the two sheet subgraphs. Write them as $2u, 2v$ with $u, v \in \{0, 1\}$. Then $d = 2(u + v - h)$ is even and at most four. □

For $d = 2$, either both visible arms belong to one sheet and $h = 0$, or they belong to different sheets and $h = 1$. For $d = 4$, necessarily $a = b = 2$ and $h = 0$. In a uniform region ($d = 0$), a hidden cut has $h = 2$; it cannot simply end there. These count incident edge germs at the point. Counting crossings of a fixed-radius circle is not equivalent, since a legal cut may enter the circle repeatedly.

Corollary 6 (An absent family)**.** *If $t_L = t_R = 0$, a mask is representable if and only if its entire visible boundary is a disjoint family of simple T–B arcs. In particular, a representable mask with a completely uniform border is uniform throughout.*

Proof. By Corollary 4, $\mathcal{B}$ is empty. There can therefore be no shared cut, and the visible boundary is exactly $\mathcal{A}$. Conversely, such a boundary gives an admissible completion with no B- or H-edges. If all four counts vanish, both families are empty and the renderer is constant. □

Example 7 (Passing local tests is not enough)**.** On a white square, put a red vertical stripe touching T and B, together with a red disk disjoint from the stripe and the boundary (Figure 2, left). Here $t_T = t_B = 2$ and $t_L = t_R = 0$; every interior boundary point has two visible arms, and all corners are white. Nevertheless the mask is impossible: Corollary 6 excludes the disk's closed boundary. An isolated motif is not universally forbidden; the absent second cut family is essential to this obstruction.



(a) Global obstruction (b) Narrow-corridor repair (c) Local obstruction

Figure 2: (a) Border counts and local degrees pass, but no completion exists. (b) A thin red corridor makes the boundary two simple top–bottom arcs: the modified mask is representable and differs by arbitrarily little area. (c) Six branches meet at the centre, violating the degree condition; the circle delimits the neighbourhood.

4 Implications for reconstruction

An obstruction is not an error bound. Keep the impossible mask in Example 7 fixed, and join its disk to its stripe with a red horizontal corridor of width $\varepsilon > 0$ (Figure 2b). For sufficiently small ε, the modified boundary consists of two simple T–B arcs, so Corollary 6 makes it representable. Only $O(\varepsilon)$ area is added. Thus even a fixed impossible target can have infimum area error zero without an exact solution. A prescribed positive paper-width constraint would obstruct this limiting construction.

Exact geometry, raster data and manufacture differ. The criterion concerns ideal two-colour regions. Blur, thresholding, cropping or a single-pixel bridge can change the interpreted boundary graph. Applying a proved obstruction to that graph does not certify the unknown design behind a photograph. Likewise, a finite ordinary polynomial-cubic Bézier model cannot represent every permissible smooth arc exactly: a nontrivial circular arc already requires approximation. Minimum widths, paper-core connectivity and cutting or threading tolerances impose further conditions beyond the theorem. A free border changes the target and its endpoint counts; it can remove border obstructions but need not repair interior junctions.

What a solver can certify. An embedded admissible completion certifies representability; a violated necessary condition certifies impossibility for exact source geometry. An infeasible restricted mixed-integer linear program only excludes its candidate graph and constraints. A timeout or stalled fit is inconclusive. A complete decision procedure must exhaust all relevant connector embeddings and attachment points along visible edges. Neither such an algorithm nor its complexity is established here.

References

[1] Torben Mogensen. *Braided Christmas Hearts*. 7 December 2021, especially Section 4, printed pages 9–10. https://hjemmesider.diku.dk/~torbenm/hearts.pdf.